

## Infinite Limits

### Example

1. Find  $\lim_{x \rightarrow 0^-} x^{-1} e^{x^{-2}}$ .

**Solution:** As  $x \rightarrow 0^+$ , we know that  $x^{-1} \rightarrow -\infty$  and  $x^{-2} \rightarrow \infty$  so  $x^{-1} e^{x^{-2}} \rightarrow -\infty \cdot e^\infty = -\infty$ .

### Problems

2. Find  $\lim_{x \rightarrow 2\pi^-} x \csc(x)$ .

**Solution:** As  $x \rightarrow 2\pi^-$ , we know that  $\sin(x) \rightarrow 0^-$  and hence  $x \csc(x) = \frac{x}{\sin(x)} \rightarrow -\infty$ .

3. Find  $\lim_{x \rightarrow 2\pi^+} x \csc(x)$ .

**Solution:** As  $x \rightarrow 2\pi^+$ , we know that  $\sin(x) \rightarrow 0^+$  and hence  $x \csc(x) = \frac{x}{\sin(x)} \rightarrow \infty$ .

4. Find  $\lim_{x \rightarrow 2^-} \frac{x^2 - 2x - 8}{-(x^2 - 5x + 6)}$ .

**Solution:** Factoring, we have that  $\frac{x^2 - 2x - 8}{-(x^2 - 5x + 6)} = \frac{(x-4)(x+2)}{-(x-2)(x-3)}$ . As  $x \rightarrow 2^-$ , we get  $\frac{-8}{0}$  and so we need to care about the sign. Since  $x < 2$ , the signs are  $\frac{- \cdot +}{- \cdot -} = +$  so the limit is  $+\infty$ .

5. Find  $\lim_{x \rightarrow 2^+} \frac{x^2 - 2x - 8}{-(x^2 - 5x + 6)}$ .

**Solution:** Factoring, we have that  $\frac{x^2 - 2x - 8}{-(x^2 - 5x + 6)} = \frac{(x - 4)(x + 2)}{-(x - 2)(x - 3)}$ . As  $x \rightarrow 2^+$ , we get  $\frac{-8}{0}$  and so we need to care about the sign. Since  $x > 2$ , the signs are  $\frac{-}{+} = -$  so the limit is  $-\infty$ .

6. Find  $\lim_{x \rightarrow 3^-} \frac{\sqrt{x}}{(x - 3)^4}$ .

**Solution:** As  $x \rightarrow 3^-$ , we have that  $(x - 3)^4 \rightarrow 0^+$  and so the limit is  $\infty$ .

7. Find  $\lim_{x \rightarrow 3^+} \frac{\sqrt{x}}{(x - 3)^4}$ .

**Solution:** As  $x \rightarrow 3^+$ , we have that  $(x - 3)^4 \rightarrow 0^+$  and so the limit is  $\infty$ .

## Implicit Differentiation

### Example

1. Find  $y'$  if  $x^3 + y^3 = 4$ .

**Solution:** Taking the derivative of both sides, we have that  $3x^2 + 3y^2 \cdot y' = 0$  and hence  $y' = \frac{-3x^2}{3y^2} = \frac{-x^2}{y^2}$ .

2. Find  $y'$  if  $e^{xy} = e^{4x} - e^{5y}$ .

**Solution:** We take the derivative to get that  $4e^{4x} - 5e^{5y}y' = e^{xy}(xy)' = e^{xy}(xy' + y)$ . Bringing all the  $y'$  to one side, we have  $4e^{4x} - ye^{xy} = 5e^{5y}y' + xe^{xy}y'$  so  $y' = \frac{4e^{4x} - ye^{xy}}{5e^{5y} + xe^{xy}}$ .

### Problems

3. Find  $y'$  if  $(x - y)^2 = x + y - 1$ .

**Solution:** Taking the derivative gives  $2(x - y)y' = 1 + y'$  so  $y' = \frac{1}{2x - 2y - 1}$ .

4. Find  $y'$  if  $y = \sin(3x + 4y)$ .

**Solution:** Taking the derivative gives  $y' = \cos(3x + 4y)(3x + 4y)' = \cos(3x + 4y)(3 + 4y')$ . Thus,  $y' = \frac{3 \cos(3x + 4y)}{1 - 4 \cos(3x + 4y)}$ .

5. Find  $y'$  if  $y = x^2y^3 + x^3y^2$ .

**Solution:** Taking the derivative gives  $y' = 2xy^3 + 3x^2y^2y' + 3x^2y^2 + 2x^3yy'$  and hence  $y' = \frac{2xy^2 + 3x^2y^2}{1 - 3x^2y^2 - 2x^3y}$ .

6. Find  $y'$  if  $\cos^2 x + \cos^2 y = \cos(2x + 2y)$ .

**Solution:** Taking the derivative gives  $-2 \cos x \sin x - 2y' \cos y \sin y = -2 \sin(2x + 2y) - 2y' \sin(2x + 2y)$  and so

$$y' = \frac{\cos x \sin x - \sin(2x + 2y)}{\sin(2x + 2y) - \cos y \sin y}.$$

7. Find  $y'$  if  $x = 3 + \sqrt{x^2 + y^2}$ .

**Solution:** Taking the derivative gives  $1 = \frac{2x + 2yy'}{2\sqrt{x^2 + y^2}}$  so solving gives

$$y' = \frac{\sqrt{x^2 + y^2} - x}{y}.$$

8. Given  $\frac{x-y^3}{y+x^2} = x + 2$ , find  $y'$ .

**Solution:** Multiply through by  $(y + x^2)$  first and take the derivative of both sides to get  $1 - 3y^2y' = xy' + y + 2y' + 3x^2 + 4x$  (you can do this without multiplying through but its a bit uglier). Then solving for  $y'$  gives

$$y' = \frac{1 - y - 3x^2 - 4x}{3y^2 + x + 2}.$$

9. Find  $y'$  give that  $\frac{y}{x^3} + \frac{x}{y^3} = x^2y^4$ .

**Solution:** Multiply through by  $x^3y^3$  to clear denominators so  $y^4 + x^4 = x^5y^7$ . Taking the derivative gives  $4y^3y' + 4x^3 = 7x^5y^6y' + 5x^4y^7$ . Solving for  $y'$  gives

$$y' = \frac{5x^4y^7 - 4x^3}{4y^3 - 7x^5y^6}.$$

10. Find  $y'$  given  $(x^2 + y^2)^3 = 8x^2y^2$ .

**Solution:** Take the derivative and solve for  $y'$  to get

$$y' = \frac{16xy^2 - 6x(x^2 + y^2)^2}{6y(x^2 + y^2)^2 - 16x^2y}.$$